1) Write a program in C to convert miles into kilometers (Km). Hint: 1 Mile=1.609 Km. [Use macros, relevant name and types for variables].

```c
#include <stdio.h>
#define mile_to_km 1.609f
int main (){
    printf("please enter the value in miles : ");
    float miles;
    scanf("%f",&miles);
    float km=miles*mile_to_km;
    printf("the value in kilometers is  : %f ",km);
return 0;
}
```

```
~/Documents/dslab  nvim assignment1_Q1.c
~/Doc/dslab  gcc assignment1_Q1.c
~/Documents/dslab  ./a.out
please enter the value in miles : 10
the value in kilometers is  : 16.090000
~/Documents/dslab  ./a.out
please enter the value in miles : 42
the value in kilometers is  : 67.577995
~/Documents/dslab  ./a.out
please enter the value in miles : 22
the value in kilometers is  : 35.397999
~/Documents/dslab  ./a.out
please enter the value in miles : 35
the value in kilometers is  : 56.314999
~/Documents/dslab  ./a.out
please enter the value in miles : 41
the value in kilometers is  : 65.969002
```

https://github.com/Udbhav-Dubey/uddsalab

Write aprogram to find the number of positive, negative and zeros in a sequence of inputs (numbers) entered as data.

```c
#include <stdio.h>
int main (){
    printf("how many numbers do you want to enter : ");
    int n;
    scanf("%d",&n);
    int positives=0,negative=0,zeros=0;
    for (int i=0;i<n;i++){
        int x;
        scanf("%d",&x);
        if (x>0){positives++;}
        else if (x==0){zeros++;}
        else negative++;
    }
    printf("the summary is \npositives : %d\nnegative : %d\nzeros : %d ",positives,negative
,zeros);
return 0;
}
```

```
~/Documents/dslab  gcc assignment1_Q2.c
~/Documents/dslab  ./a.out
how many numbers do you want to enter : 6
1
2
-1
-3
0
5
the summary is
positives : 3
negative : 2
zeros : 1
                                    ✓ 13s
how many numbers do you want to enter : 11
1
2
3

-1
2
0
--2
112
-5
-7
the summary is
positives : 5
negative : 4
zeros : 2
```

3) Compute the tax due based on a tax table given below:
Program Input: Salary amount.
Program Output: Returns the tax due for 0.0 <= salary <= 150,000.00; returns -1.0 if salary is outside the table range.

Salary Range ($)	Base Tax ($)	Percentage of Excess
0.00–14,999.99	0.00	15
15,000.00–29,999.99	2,250.00	18
30,000.00–49,999.99	5,400.00	22
50,000.00–79,999.99	11,000.00	27
80,000.00–150,000.00	21,600.00	33

```c
#include <stdio.h>
float taxforsalary(float salary){
    if(salary <0.0f || salary >150000.00f){
        return -1.0;
    }
    float tax;
    if (salary<14999.99f){
        tax=0.00f+(salary-0.00f)*0.15f;
    }
     else if (salary<29999.99f){
        tax=2250.00f+(salary-15000.00f)*0.18f;
    }
     else if (salary <= 49999.99f) {
        tax = 5400.00f + (salary - 30000.00f) * 0.22f;
    }
     else if (salary <= 79999.99f) {
        tax = 11000.00f + (salary - 50000.00f) * 0.27f;
    }
     else {
        tax = 21600.00f + (salary - 80000.00f) * 0.33f;
    }
    return tax;
}
int main (){
    float salary;
    printf("enter the amount of salary : ");
    scanf("%f",&salary);
    if (salary<0.0f){
        printf("invalid input\n");
    }
    else{
        float tax=taxforsalary(salary);
        printf("the tax due is : %f",tax);
    }

    return 0;
}
```

ORMAL assignment1_03.c

```
~/Documents/dslab  gcc assignment1_Q3.c
~/Documents/dslab  ./a.out
enter the amount of salary : 12000
the tax due is : 1800.000122
~/Documents/dslab  ./a.out
enter the amount of salary : 16550
the tax due is : 2529.000000
~/Documents/dslab  ./a.out
enter the amount of salary : -1400
invalid input
~/Documents/dslab  ./a.out
enter the amount of salary : 1555000
the tax due is : -1.000000
~/Documents/dslab  ./a.out
enter the amount of salary : 11
the tax due is : 1.650000
~/Documents/dslab  ./a.out
enter the amount of salary : 1212121
the tax due is : -1.000000
~/Documents/dslab  ./a.out
enter the amount of salary : 10000
the tax due is : 1500.000000
~/Documents/dslab
```

4) Write an interactive program (menu driven) in ‘C’ (using functions) to compute the area of a selected geometrical figure from a list of such figures (square, rectangle, and circle).

```c
#include <stdio.h>
#include <stdlib.h>
#define pie 3.14f
int main (){
    while(true){
    printf("please select the figure by typing the number against it : \n");
    printf("1.sqaure\n2.rectange\n3.circle\n4.exit\n");
    int n;
    printf("now enter the number : ");
    scanf("%d",&n);
    if (n==1){
        printf("you have selected a square now enter the length of side for it : ");
        float s;
        scanf("%f",&s);
        printf("the area is %f \n",s*s);
    }
    else if (n==2){
        printf("you have selected a rectangle now enter the dimensions\n");
        printf("enter the length : ");
        float l;
        scanf("%f",&l);
        float b;
        printf("enter the breadth : ");
        scanf("%f",&b);
        printf("the area is %f\n",l*b);
    }
    else if (n==3){
        printf("you have selected a circle now enter its radius : ");
        float r;
        scanf("%f",&r);
        printf("the area is %f\n",pie*r*r);
    }
    else if (n==4){
        printf("thanks for using the program\n");
        break;
        }
    else {printf("invalid number \n");
        break;}}
return 0;
}
```

```
~/Documents/dslab  ./a.out

please select the figure by typing the number against it :
1.sqaure
2.rectange
3.circle
4.exit
now enter the number : 1
you have selected a square now enter the length of side for it : 22
the area is 484.000000

please select the figure by typing the number against it :
1.sqaure
2.rectange
3.circle
4.exit
now enter the number : 2
you have selected a rectangle now enter the dimensions
enter the length : 1.5
enter the breadth : 2.3
the area is 3.450000

please select the figure by typing the number against it :
1.sqaure
2.rectange
3.circle
4.exit
now enter the number : 3
you have selected a circle now enter its radius : 5.6
the area is 98.470398

please select the figure by typing the number against it :
1.sqaure
2.rectange
3.circle
4.exit
now enter the number : 4
thanks for using the program
```

5) Write a program to display first n elements of Fibonacci series.

```c
//Write a program to display first n elements of Fibonacci series.
#include <stdio.h>
int main (){
    printf("enter the nth element to find fibonacci : ");
    int n;
    scanf("%d",&n);
    int a=0;
    int b=1;
    int c;
    for (int i=0;i<n;i++){
        printf("%d  ",a);
        c=a+b;
        a=b;
        b=c;
    }
return 0;
}
```

```
~/Doc/dslab  gcc assignment1_Q5.c                                  ✔  29s
~/Documents/dslab  ./a.out                                          ✔
enter the nth element to find fibonacci : 5
0  1  1  2  3
~/Documents/dslab  ./a.out                                          ✔
enter the nth element to find fibonacci : 11
0  1  1  2  3  5  8  13  21  34  55
~/Documents/dslab  ./a.out                                          ✔
enter the nth element to find fibonacci : 33
0  1  1  2  3  5  8  13  21  34  55  89  144  233  377  610  987  1597  2
~/Documents/dslab  ./a.out                                          ✔
enter the nth element to find fibonacci : 13
0  1  1  2  3  5  8  13  21  34  55  89  144
~/Documents/dslab  ./a.out                                          ✔
```

6) Write a program to print a table book from Table X to Table Y. X and Y are user inputs.

```c
#include <stdio.h>
int main (){
    printf("enter the starting table : ");
    int x;
    scanf("%d",&x);
    printf("enter the ending table : ");
    int y;
    scanf("%d",&y);
    printf("table book from %d to %d ---\n\n",x,y);
    for (int i=x;i<=y;i++){
        printf("table of %d : \n",i);
        for (int j=1;j<=10;j++){
            printf("%d*%d=%d\n",i,j,i*j);
        }
        printf("\n");
    }
return 0;
}
```

```
~/Documents/dslab  ./a.out
enter the starting table : 4
enter the ending table : 6
table book from 4 to 6 ---

table of 4 :
4*1=4
4*2=8
4*3=12
4*4=16
4*5=20
4*6=24
4*7=28
4*8=32
4*9=36
4*10=40

table of 5 :
5*1=5
5*2=10
5*3=15
5*4=20
5*5=25
5*6=30
5*7=35
5*8=40
5*9=45
5*10=50

table of 6 :
6*1=6
6*2=12
6*3=18
6*4=24
6*5=30
6*6=36
6*7=42
6*8=48
6*9=54
6*10=60

~/Documents/dslab
```

7) Write a program to compute factorial of a number using iterative approach.

```
//Write a program to compute factorial of a number using iterative approach.
#include <stdio.h>
int main (){
    long long int n;
    printf("please enter the number : ");
    scanf("%lld",&n);
    long long result=1;
    for (long long int i=1;i<=n;i++){
        result*=i;
    }
    printf("the answer is %lld",result);
return 0;
}
```

```
zsh: no such file or directory: ./run/me
~/Documents/dslab ./a.out
please enter the number : 5
the answer is 120
~/Documents/dslab ./a.out
please enter the number : 6
the answer is 720
~/Documents/dslab ./a.out
please enter the number : 1
the answer is 1
~/Documents/dslab ./a.out
please enter the number : 0
the answer is 1
~/Documents/dslab ./a.out
please enter the number : 99
the answer is 0
~/Documents/dslab ./a.out
please enter the number : 20
the answer is 2432902008176640000
~/Documents/dslab
```

8) Write a program to swap two numbers using functions.

```c
#include <stdio.h>
int main (){
    float n1;
    float n2;
    printf("enter the first number : ");
    scanf("%f",&n1);
    printf("enter the second number : ");
    scanf("%f",&n2);
    float temp;
    temp=n1;
    n1=n2;
    n2=temp;
    printf("now the first number is %.2f  \n",n1);
    printf("now the second number is %.2f  \n",n2);
return 0;
}
```

```
~/Documents/dslab  nvim assignment1_Q8.c
~/Documents/dslab  gcc assignment1_Q8.c
~/Documents/dslab  ./a.out
enter the first number : 2.6
enter the second number : 4.8
now the first number is 4.80
now the second number is 2.60
~/Documents/dslab  ./a.out
enter the first number : 0.00052
enter the second number : .002
now the first number is 0.00
now the second number is 0.00
~/Documents/dslab  ./a.out
enter the first number : 112.22
enter the second number : 12321.12
now the first number is 12321.12
now the second number is 112.22
```

9) Write a function that returns the first integer between n_min and n_max entered as data to the calling function (main).

```c
#include <stdio.h>
int firstint(int n_min,int n_max){
    return n_min;
}
int main (){
    printf("enter the range\n");
    printf("starting point : ");
    int n_min;
    scanf("%d",&n_min);
    printf("ending point : ");
    int n_max;
    scanf("%d",&n_max);
    if (n_min>n_max){printf("wrong range ");return 0;}
    else {printf("the answer is %d",firstint(n_min,n_max));}
return 0;
}
```

```
wrong range
~/Documents/dslab ./a.out
enter the range
starting point : 10
ending point : 100
the answer is 10
~/Documents/dslab ./a.out
enter the range
starting point : 12
ending point : 15
the answer is 12
~/Documents/dslab ./a.out
enter the range
starting point : 16
ending point : 11
wrong range
```

10) Write nests of loops that cause the following output to be displayed.

```
0
0 1
0 1 2
0 1 2 3
0 1 2 3 4
0 1 2 3 4 5
0 1 2 3 4
0 1 2 3
0 1 2
```

```c
#include <stdio.h>
int main (){
for (int i=0;i<6;i++){
    for(int j=0;j<=i;j++){
        printf("%d ",j);
        }
        printf("\n");
    }
for (int i=4;i>=0;i--){
    for (int j=0;j<=i;j++){
        printf("%d ",j);
        }
        printf("\n");
    }
return 0;
}
```

```
~/Documents/dslab  gcc assignment1_Q10.c
~/Documents/dslab  ./a.out
0
0 1
0 1 2
0 1 2 3
0 1 2 3 4
0 1 2 3 4 5
0 1 2 3 4
0 1 2 3
0 1 2
0 1
0
```